

Haoyu Zhang

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Education

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| MS Southwest University , Operations Research
• Top 5% (3/65): Optimization, Variational Analysis, etc. | Chongqing, China
Sept 2025 – present |
| BS Donghua University , Mathematics and Applied Mathematics
• Top 14% (8/61): Mathematical Modeling, Discrete Mathematics, Data Structures and Algorithms, etc. | Shanghai, China
Sept 2021 – June 2025 |

Experience

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| Appen Ltd. , Math AIGC Training Intern
• Designed Chain-of-Thought (CoT) structures to improve the logical consistency and interpretability of Large Language Models (LLMs) in solving complex mathematical problems.
• Participated in establishing the company's first set of data annotation specifications for mathematical reasoning chains, achieving structured reasoning paths and unified annotation standards. | Chongqing, China
Jan 2025 – Apr 2025
4 months |
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Research Experience

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| Fluid Dynamics and Thermal Coupling Modeling in Greenhouse Environments
Group Leader
• Established 3D wind speed and temperature distribution models for glass greenhouses based on Navier-Stokes (N-S) and heat conduction equations, accounting for heat loss from plant transpiration and correcting for complex airflow in crop scenarios.
• Developed a numerical solver using finite difference and control volume methods to discretize and solve complex PDEs and visualize distribution results, providing a modeling basis for intelligent agricultural environment control. | Nov 2023 – Apr 2024 |
| Mathematical Modeling and Research of Mixed Human-Machine Traffic Flow
Independent Researcher (Advisor: Prof. Linglong Du)
• Developed a mathematical model for mixed traffic flow of human-driven and autonomous vehicles in ring-road scenarios, incorporating the Bando-Follow the Leader (Bando-FTL) model and accounting for stochastic slowdown effects.
• Introduced a Proportional-Integral (PI) controller for Autonomous Vehicles (AVs) to optimize regulation capabilities. | Sept 2024 – Apr 2025 |

Projects

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| Climbing Posture Analysis and Coordination Optimization Based on Computer Vision
• Utilized OpenPose for precise recognition and tracking of human key points across three types of climbing scenarios.
• Designed a frame counter to calculate displacement via discrete accumulation, visualized motion curves, and computed a synchronization index through phase domain transformation to evaluate athlete coordination. | Sept 2023 – Jan 2024 |
| Nighttime Vehicle Speed Estimation Model Based on Optical Flow Analysis
• Implemented an algorithm for motion feature extraction based on Lucas-Kanade optical flow to identify vehicle contour lights.
• Analyzed the impact of optical flow thresholds on noise filtering and target recognition stability for intelligent traffic detection. | Mar 2024 – June 2024 |

Skills

Expertise: Distributed Optimization & Primal-Dual Algorithms, Multi-agent System Modeling, Numerical Solution and Simulation of PDEs, Control System Design (PID/PI)

Languages: Mandarin (Native), English (CET-6), Japanese (JLPT N2)

Programming: Python (Pandas, PyTorch, Sklearn, OpenCV), MATLAB, Java, LaTeX/Markdown

Honors and Awards

- First-Prize Graduate Academic Scholarship, Southwest University (2025)
- Tianji Annual Scholarship, Donghua University (2024)
- Honorable Mention, MCM/ICM (2023)
- Third-Prize Undergraduate Academic Scholarship, Donghua University (2022, 2023)
- Third Prize (Shanghai Division), Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) (2023)
- Third Prize, Asia-Pacific Mathematical Contest in Modeling (APMCM) (2023)